

The text discusses Verification and Validation (V&V) in software requirements and design specifications. Its main purpose is to explain how software problems and high-risk issues can be identified and resolved as early as possible in the software development life cycle. According to the paper, detecting and correcting problems early can generate significant cost savings compared with fixing them in later stages. Early V&V can also improve the reliability, maintainability, and overall quality of the final software products.

The paper distinguishes between verification and validation, two concepts that are closely related but have different purposes. Verification checks whether a software corresponds correctly to its specifications.

In other words, it asks: "Are we building the product right?" Validation focuses on whether the software is suitable for its intended operational purpose and whether it actually satisfies the needs of users and customers. Therefore, validation asks: "Are we building the right product?"

These activities take place throughout the software life cycle. Verification checks whether the different stages of development follow the original requirements, while validation determines whether the requirements and the final product meet the actual needs of users. The paper identifies four main criteria for evaluating software specifications: Completeness, consistency, feasibility, and testability.

Complete specifications includes all necessary requirements and goals. Consistency ensures that requirements do not conflict with each other. Feasibility requires requirements to be clear, specific, and measurable.